

LUHANSK INTL, Ukraine

WMO: 330001

Lat: 48.417N

Lon: 39.374E

Elev: 636

StdP: 14.36

Time Zone: 2.00 (EUE)

Period: 04-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.5	-2.2	-12.6	2.8	-7.3	-6.2	4.0	-1.3	28.6	24.3	26.1	26.1	7.4	60	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.6	95.1	66.3	91.5	66.0	87.9	65.2	70.2	83.3	69.1	82.6	67.8	81.5	12.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	99.3	74.4	64.7	93.5	73.2	63.0	88.1	72.3	34.5	84.5	33.5	82.2	32.5	81.4	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.5	22.3	20.1	-10.2	99.0	7.5	3.1	-15.6	101.2	-20.0	103.0	-24.2	104.8	-29.6	107.0		
(5)			-10.5	73.6	7.4	1.9	-15.9	75.0	-20.3	76.1	-24.4	77.2	-29.8	78.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.9	22.4	22.8	34.4	49.8	62.9	69.1	73.0	73.2	61.2	49.0	38.0	28.9		
	DBStd	20.27	12.94	12.30	8.31	7.58	7.82	6.22	6.06	7.16	6.67	7.92	8.36	9.73		
	HDD50	3350	855	762	487	94	7	0	0	0	4	116	370	655		
	HDD65	6665	1320	1182	950	457	135	28	7	13	148	495	811	1119		
	CDD50	2937	0	0	2	89	406	573	713	718	341	86	9	0		
	CDD65	780	0	0	0	2	69	151	255	267	35	1	0	0		
	CDH74	8520	0	0	0	37	773	1568	2680	3083	372	7	0	0		
	CDH80	3508	0	0	0	4	230	561	1154	1471	88	0	0	0		
	WSAvg	9.4	10.9	11.1	11.6	9.9	8.3	7.8	7.2	7.7	8.4	8.9	9.7	11.3		
	PrecAvg	21.0	1.6	1.5	1.5	1.3	1.8	2.4	2.4	1.3	1.9	1.6	1.6	2.0		
(15) Precipitation	PrecMax	29.1	3.4	3.7	2.9	2.7	4.1	5.2	6.0	3.1	5.0	3.6	3.0	4.5		
	PrecMin	14.3	0.6	0.5	0.7	0.2	0.2	0.6	0.2	0.2	0.1	0.6	0.1	0.8		
	PrecStd	4.3	0.8	0.7	0.6	0.6	0.9	1.3	1.5	0.7	1.3	0.8	0.7	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.7	44.5	61.2	79.1	91.3	93.5	98.4	101.9	87.6	73.4	62.7	51.8		
		MCWB	44.6	39.9	48.9	57.7	65.5	66.7	65.9	66.3	61.9	57.0	53.9	50.9		
	2%	DB	42.7	41.0	55.1	71.9	85.8	89.9	93.6	97.0	82.3	68.4	57.1	46.6		
		MCWB	41.2	39.1	45.5	55.4	63.7	66.4	66.7	66.1	62.9	56.9	50.9	45.1		
	5%	DB	39.5	38.8	50.0	67.8	82.2	86.2	90.0	93.0	78.5	64.6	52.1	43.2		
		MCWB	38.4	37.6	43.7	53.3	62.5	65.0	67.0	65.2	61.0	55.9	48.4	42.2		
	10%	DB	37.1	37.0	46.3	63.0	78.5	82.5	86.3	88.0	74.9	60.9	49.6	40.8		
		MCWB	36.3	36.0	42.1	51.0	60.8	64.7	66.2	64.7	59.3	55.5	47.0	40.3		
	0.4%	WB	44.8	42.0	51.0	60.1	68.9	71.5	73.3	70.8	68.2	62.8	54.6	51.1		
		MCDB	46.1	42.7	58.3	75.3	84.6	82.8	89.0	87.1	78.3	69.8	60.3	52.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.1	39.1	47.4	57.2	66.0	69.5	70.8	69.2	65.3	59.8	51.6	44.9		
		MCDB	41.7	40.2	52.7	69.8	80.1	82.4	82.9	84.8	75.5	65.3	56.2	45.8		
	5%	WB	38.3	37.3	44.5	54.9	64.4	68.0	69.6	67.9	63.1	57.7	49.5	42.4		
		MCDB	39.3	38.0	48.2	64.1	77.0	81.6	82.1	83.7	73.1	62.3	52.1	43.1		
	10%	WB	35.7	35.5	42.0	52.8	62.6	66.7	68.2	66.7	61.2	55.7	46.8	39.7		
		MCDB	36.6	36.2	45.4	60.9	74.8	79.2	80.8	82.6	70.2	59.8	48.8	40.3		
	5% DB	MDBR	10.0	10.8	14.0	19.3	21.5	21.4	21.7	23.6	20.9	15.4	11.2	8.2		
		MCDBR	9.4	10.9	20.7	25.5	26.5	26.0	26.2	28.7	26.1	20.0	16.2	9.5		
		MCWBR	8.4	9.1	13.3	13.2	11.1	10.0	9.4	9.8	11.5	11.4	11.1	8.4		
		MCWBR	9.2	8.9	18.0	21.3	22.6	22.0	21.5	23.2	21.4	17.2	13.4	9.4		
(35) Mean Daily Temperature Range	5% WB	MCWBR	8.6	7.9	12.7	11.9	10.8	9.8	9.3	9.4	11.1	10.9	10.1	8.7		
	Clear-Sky Solar Irradiance	taub	0.327	0.358	0.390	0.448	0.413	0.411	0.435	0.436	0.406	0.374	0.347	0.331		
		taud	2.194	2.115	2.158	2.071	2.262	2.298	2.241	2.220	2.269	2.322	2.305	2.231		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	224	243	257	253	270	271	261	255	250	239	217	206		
		Edh at Noon	26	35	40	48	41	40	42	41	36	29	23	22		
(44) All-Sky Solar Radiation	RadStd		313	611	943	1378	1846	1994	1948	1778	1241	706	331	203		
			71	126	148	148	164	167	123	122	120	106	65	29		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		+1.22	N/A	N/A	N/A	N/A	N/A	N/A	-311	+211	+133

Nomenclature: See separate page